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Key Points:

- A hybrid approach is proposed using a Random Forest-corrected Noah land surface model (LSM) to define lower boundary for the physically-advanced Simultaneous Heat and Water (SHAW) model
- The hybrid approach significantly outperforms standalone Noah LSM and traditional SHAW configurations in simulating active layer thermo-hydrology
- Correcting Noah LSM simulations with machine learning offers a transferable method for constraining other permafrost models in data-sparse region

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A Hybrid Modeling Approach for Improved Simulation of Thermal-Hydrological Dynamics in Active Layer on the Qinghai-Tibet Plateau

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Abstract Accurately modeling active layer dynamics on the warming and wetting Qinghai-Tibet Plateau (QTP) is crucial for understanding local hydrological processes, ecosystem dynamics, and infrastructure integrity. However, land surface models (LSMs) are often limited by simplified representation of physical processes in frozen ground and by uncertain forcing data. We present a novel hybrid modeling approach that enables the use of the Simultaneous Heat and Water (SHAW) model, with its advanced representation of physical processes in frozen ground, in data-sparse regions by generating robust lower boundary conditions from random forest-corrected Noah LSM simulations. When evaluated at seven permafrost sites on the QTP, the hybrid approach significantly outperformed both the standalone Noah LSM and traditional SHAW configurations in simulating active layer temperature and moisture. For testing data, the hybrid approach achieved higher average Nash-Sutcliffe efficiency values for soil temperature (ST) (0.81 vs. 0.69) and soil moisture (0.35 vs. 0.17) compared to the Noah LSM. Notably, the hybrid approach corrects key biases of the Noah LSM, which tends to overestimate ST and unfrozen water content during freezing periods. This study provides a robust framework for large-scale simulation of permafrost dynamics in data-sparse regions, with direct implications for assessing environment change, infrastructure risks, and carbon emissions linked to permafrost degradation.

Plain Language Summary Permafrost is ground that remains frozen ($\leq 0^{\circ}\text{C}$) for at least two consecutive years. The Qinghai-Tibet Plateau holds the world's largest area of high-altitude permafrost. As the climate warms, this permafrost is thawing, which threatens regional water resources, ecosystem, and infrastructure. Current computer models often fail to accurately predict this thaw because they oversimplify the complex processes in frozen soil. This study's hybrid solution makes an advanced model that excels at simulating the physics of frozen ground, usable in remote regions. We accomplish this by generating the temperature deep underground using a second model (Noah) combined with machine learning. When tested at seven sites, our approach was significantly better at predicting soil temperature (ST) and water content than traditional models. In particular, it corrected common errors where other models overestimate ST and unfrozen water content during winter. This approach provides a more reliable way to study large-scale permafrost changes, helping researchers better assess environmental impacts, infrastructure risks, and carbon release from thawing ground.

1. Introduction

The Qinghai-Tibet Plateau (QTP) holds the world's largest high-altitude permafrost region, encompassing approximately $1.1 \times 10^6 \text{ km}^2$ (Cao et al., 2023). Air temperatures on the QTP have risen at a rate (0.32°C per decade) twice the global average (0.16°C per decade) (Pepin et al., 2015; Smith et al., 2022), leading to substantial permafrost degradation (Wu et al., 2018; Zhang et al., 2022). This warming has profound consequences, impacting regional water availability (Li et al., 2021; Sjöberg et al., 2021; Yao et al., 2022), ecosystems (Ehlers et al., 2022; Hao et al., 2021; Wang et al., 2012), the carbon cycle (Chen et al., 2022; Mu et al., 2020), and infrastructure stability (Hjort et al., 2022; Li et al., 2022; Ran et al., 2022).

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The active layer is the near-surface layer of ground that undergoes seasonal thawing and freezing in permafrost regions (Dobiński, 2020). As the critical interface between the atmosphere and permafrost, the active layer is highly sensitive to climate change, regulating the heat and water transported into the underlying permafrost (Dobiński, 2020; Smith et al., 2022). Climate warming and wetting have significantly altered the thermo-hydrological state of the QTP active layer, profoundly impacting local water cycles and vegetation growth (Del Vecchio et al., 2024; Li, Wu, et al., 2024; Liljedahl et al., 2024; Walvoord & Kurylyk, 2016). Seasonal freeze-thaw cycles in the active layer induce ground uplift and subsidence that threaten infrastructure and can accelerate the release of permafrost carbon (Hjort et al., 2022; Schuur et al., 2015; Torn et al., 2025; Yi et al., 2024). A comprehensive understanding of these thermo-hydrological variations is therefore crucial. However, current understanding relies on sparse in situ data (Chen et al., 2021; Ma et al., 2020; Zhao et al., 2021), which are insufficient to capture the spatial variability of active layer dynamics across the QTP.

Land surface models (LSMs) that incorporate soil freeze-thaw processes have emerged as valuable tools for investigating active layer dynamics at large spatial scales. While numerous LSMs have been employed to study the QTP, their accuracy is often limited by inadequate representation of freeze-thaw processes and a lack of high-quality forcing data (Hu et al., 2020; Walvoord & Kurylyk, 2016; Wu et al., 2018). For instance, the Common Land Model has been reported to significantly underestimate unfrozen water content during frozen periods (Xiao et al., 2013). Similarly, the Noah LSM exhibits biases in simulating deep active layer temperatures and seasonal soil moisture variations (Wu et al., 2018; Zheng et al., 2017). These limitations highlight an urgent need for advanced modeling approaches that incorporate more comprehensive soil heat and water processes to improve thermo-hydrological simulations within the active layer.

The Simultaneous Heat and Water (SHAW) model (Flerchinger & Saxton, 1989) is particularly well-suited for this purpose. The SHAW model incorporates a detailed representation of coupled heat and water interactions within frozen soil, including conductive, latent, and convective heat transfer (Alkhaier et al., 2012; Cui et al., 2020; Li, Wang, & Chen, 2024; Marshall et al., 2021; Zhao et al., 2022). However, the SHAW model's performance is highly sensitive to its lower boundary conditions (Alkhaier et al., 2012; Flerchinger, 1991; Zhao et al., 2023), a dependency that limits its application in data-sparse regions where accurate lower boundary data are often unavailable. In contrast, the Noah LSM, despite its own limitations, can produce reasonable simulations of deep soil states without requiring precise lower boundary constraints (Ji et al., 2022; Wu et al., 2018; Zhang et al., 2019).

Recognizing these complementary attributes, this study proposes a novel hybrid approach that integrates the SHAW model and the Noah LSM. The approach leverages Noah LSM simulations for deep soil temperatures, corrected by a Random Forest (RF) model, to generate robust lower boundary conditions for the SHAW model. This hybrid framework enables the physically-advanced SHAW model to accurately simulate thermal and hydrological processes within the active layer of the QTP permafrost regions.

2. Methods

2.1. Study Area and Data

The performance of the proposed hybrid model was evaluated using a synthesis data set from seven permafrost monitoring sites on the QTP (Zhao et al., 2021). These sites are situated along the Qinghai-Tibetan Highway and span altitudes from 4,538 m to 4,804 m and latitudes from 31.82° N to 35.72° N (Figure 1). Table 1 provides detailed site information, including location, altitude, observation periods, and active layer thickness (ALT). Each site is equipped with automated instruments that continuously record ST and moisture content at multiple depths within the active layer.

Meteorological forcing data for the models were sourced from the China Meteorological Forcing Data set (ITP-forcing), which provides data from 1979 to 2018 at a spatial resolution of 0.1° and a temporal resolution of 3 hr (He et al., 2020). Importantly, to ensure the broad applicability of our method across the QTP, site-specific meteorological observations were not used to drive the models. The ITP-forcing data set has demonstrated high accuracy in previous permafrost modeling studies on the QTP (Deng et al., 2020; Hu et al., 2022; Yang et al., 2021; Zhang et al., 2022) and is therefore considered a reliable data source for this work. We evaluated the accuracy of the ITP-forcing data set against six meteorological stations on the QTP (Figure 1a). The air temperature data show strong agreement with observations, with coefficients of determination (R^2) consistently

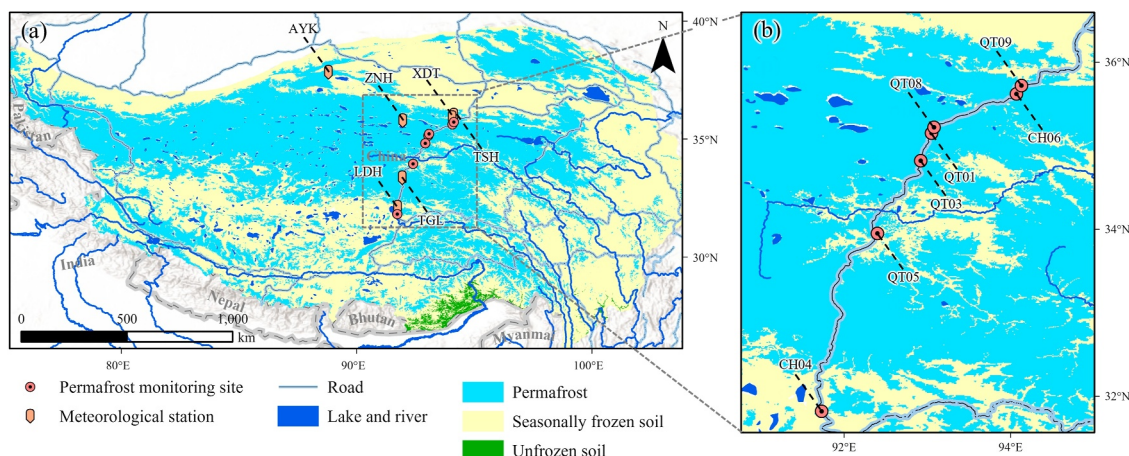


Figure 1. Distribution of permafrost and the observation sites on the Qinghai-Tibet Plateau (QTP). (a) Permafrost extent across the QTP based on Cao et al. (2023) and the locations of six meteorological stations used to evaluate the ITP-forcing data set. (b) Locations of seven permafrost monitoring sites used to assess the performance of the proposed hybrid modeling framework.

exceeding 0.90 across all stations (Figure 2a). While the ITP-forcing precipitation data captures seasonal patterns and annual totals effectively, some discrepancies were observed during peak summer rainfall events, resulting in lower R^2 values (Figure 2b). We attribute these mismatches to the high spatial heterogeneity of precipitation in mountain regions, where point-based station measurements may not represent the areal average of a 0.1° grid cell, especially given substantial elevation differences. Vegetation type for each site was derived from MODIS MCD12Q1 land cover data with a resolution of 500 m, using classes defined by the International Geosphere-Biosphere Program (Loveland & Belward, 1997). Soil texture for each site was determined using the China Soil Map based on the Harmonized World Soil Database (HWSD) (Fischer et al., 2008). Both data sets were resampled to a spatial resolution of 0.1° to align with the meteorological forcing data.

2.2. A Hybrid Modeling Approach

2.2.1. Framework of the Hybrid Modeling Approach

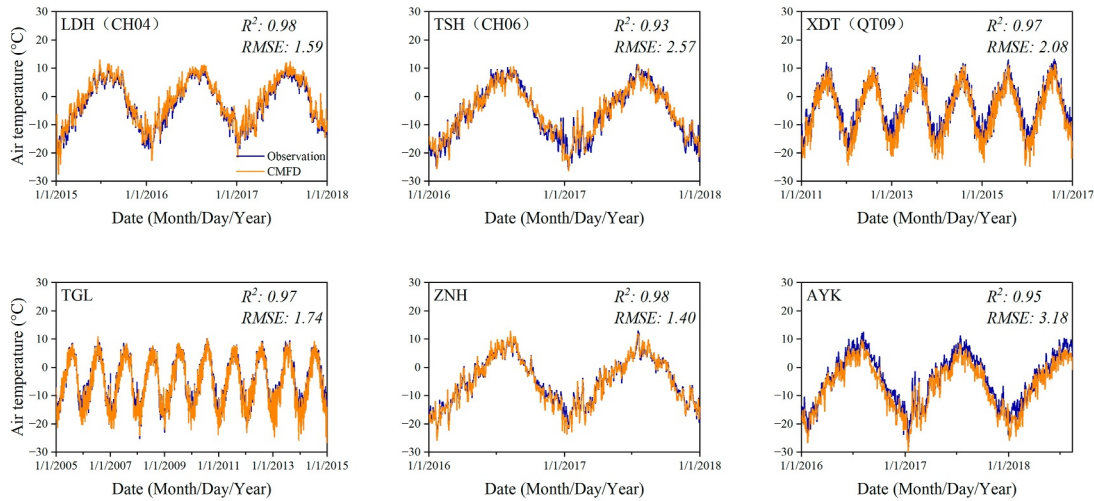
Figure 3 illustrates the framework of the hybrid modeling approach. The SHAW model serves as the primary tool to simulate active layer thermo-hydrological dynamics. Meteorological forcing data for the SHAW model were provided by the ITP-forcing data set. A key innovation is the generation of high-quality lower boundary condition to accurately regulate heat and water exchanges at the base of the active layer. To achieve this, Noah LSM simulations are corrected by an RF model. Site-specific soil and vegetation characteristics were extracted from the HWSD and MODIS data sets, respectively.

Table 1
Information of the Seven Permafrost Monitoring Sites on the Qinghai Tibet Plateau (QTP)

Station ID	Location	Longitude ($^\circ$ E)	Latitude ($^\circ$ N)	Altitude (m)	ALT (m)	Observation periods	Observation depths (m)	
							ST	SWC
CH04	Liangdaohe	91.73	31.82	4,808	1.20	2002–2003; 2005–2006	0.15, 0.5, 0.95, 1.2	0.15, 0.5, 0.95, 1.2
CH06	Kunlun pass	94.07	35.62	4,746	1.46	2013–2014	0.1, 0.5, 1.0, 1.8	0.1, 0.4, 1.0, 1.8
QT01	Kekexili	93.05	35.15	4,734	1.76	2004–2013	0.1, 0.5, 1.0, 1.8	0.1, 0.5, 1.0, 1.8
QT03	Beiluhe	92.92	34.82	4,656	2.41	2006–2014	0.1, 0.5, 0.9, 2.0	0.05, 0.5, 0.9, 1.8
QT05	Kaixinglin	92.4	33.95	4,652	3.08	2004–2008	0.1, 0.5, 0.9, 2.1	0.1, 0.5, 0.9, 2.1
QT08	Wudaoliang	93.08	35.22	4,783	2.28	2013–2018	0.1, 0.4, 1.2, 2.0	0.1, 0.4, 1.2, 2.0
QT09	Xidatan	94.13	35.72	4,538	1.37	2011–2018	0.1, 0.4, 1.2, 1.6	0.1, 0.4, 1.2, 1.6

Note. ST: soil temperature; SWC: soil water content.

(a) Air temperature evaluation



(b) Precipitation evaluation

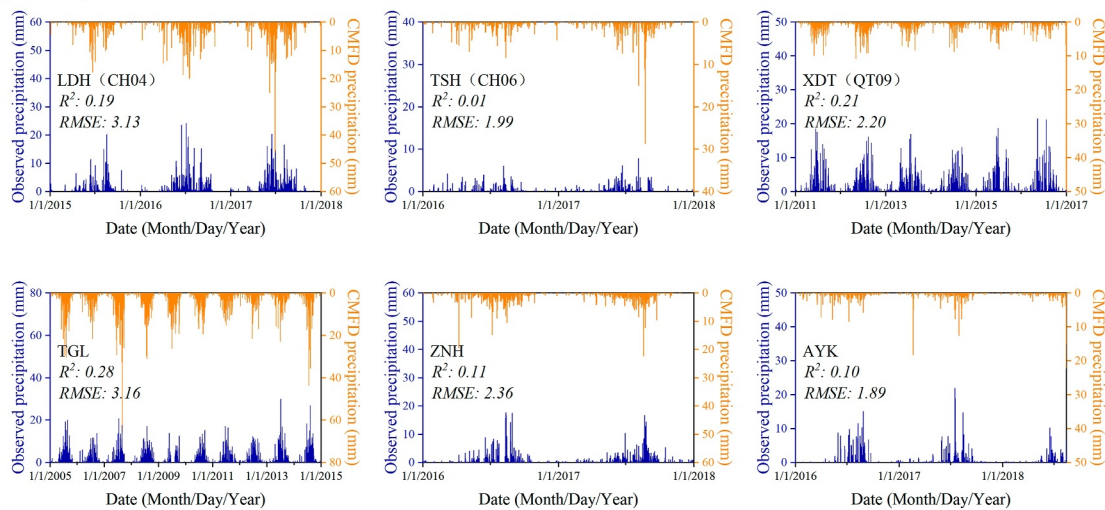


Figure 2. Comparison of ITP-forcing air temperature (a) and precipitation (b) data using in situ observations from six meteorological stations in the permafrost regions of the Qinghai-Tibet Plateau. Station locations are shown in Figure 1, and three of these stations (LDH, TSH, and XDT) are adjacent to permafrost sites evaluated in this study.

2.2.2. The Simultaneous Heat and Water Model

The SHAW model was developed to simulate heat, water, and solute transfer within a one-dimensional plant-snow-soil system (Flerchinger & Saxton, 1989). Differentiating itself from models like Noah-MP and Community Land Model (CLM), the SHAW model incorporates a more comprehensive representation of heat and water transfer processes within frozen soil, explicitly accounting for convective heat transfer and vapor migration (Flerchinger, 2000; Yu et al., 2018). As a result, the SHAW model has demonstrated superior performance in representing the coupled heat and water interactions within the permafrost active layer compared to other LSMs (Hu et al., 2020; Zhao et al., 2022, 2023).

In our previous work (Zhao et al., 2023), we enhanced the SHAW model by incorporating a modified soil matric potential parameterization that accounts for the influence of pore ice on hydrological processes in frozen ground. This improvement significantly enhances the simulation of ice/liquid water content and ST during freeze-thaw cycles. However, we also observed that the SHAW model's performance is highly sensitive to its lower boundary condition, a dependency that has been documented in previous studies (Alkhaier et al., 2012; Marshall

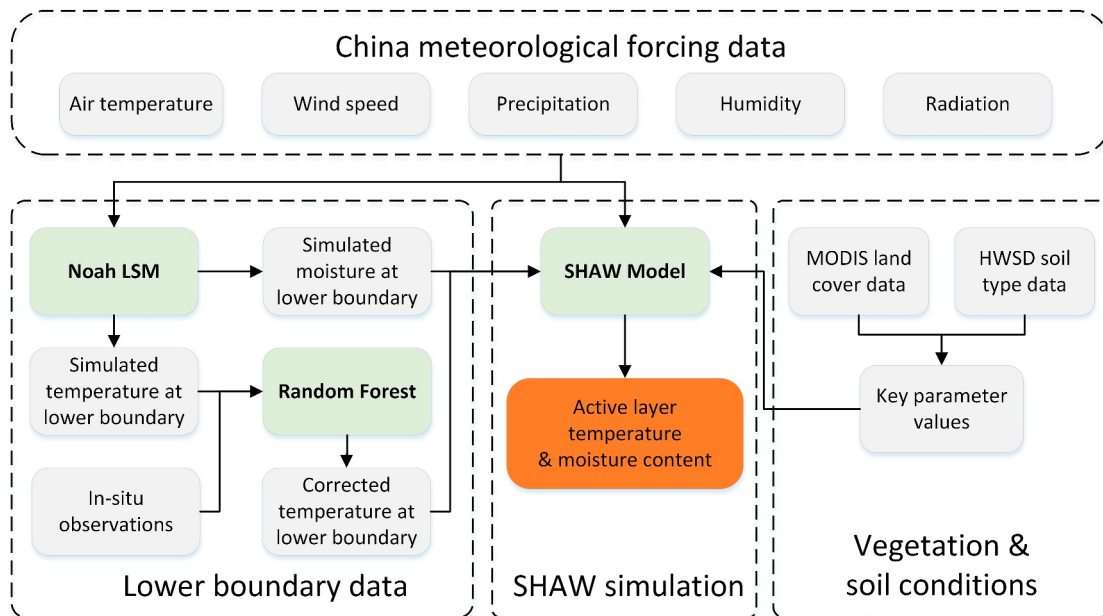


Figure 3. Framework of the hybrid modeling approach simulating the active layer temperature and moisture dynamics in the Qinghai-Tibet Plateau permafrost regions. HWSD: Harmonized World Soil Database; SHAW: the Simultaneous Heat and Water model.

et al., 2021) and confirmed in our own work. Traditional LSMs often address lower boundary uncertainty by setting a constant temperature at a great depth, minimizing its influence on upper layer simulations. However, when this approach is used with SHAW, it leads to a substantial underestimation of seasonal temperature fluctuations at the deep active layer (Figure 4c), compared to simulations using observed temperatures at 2.8 m as the lower boundary (Figure 4f). Parameter sensitivity analysis indicated that these biases do not stem from parameter uncertainty, but rather from the boundary specification itself. This limitation significantly constrained the application of the SHAW model across the QTP, where accurate, deep ST for boundary conditions are generally unavailable.

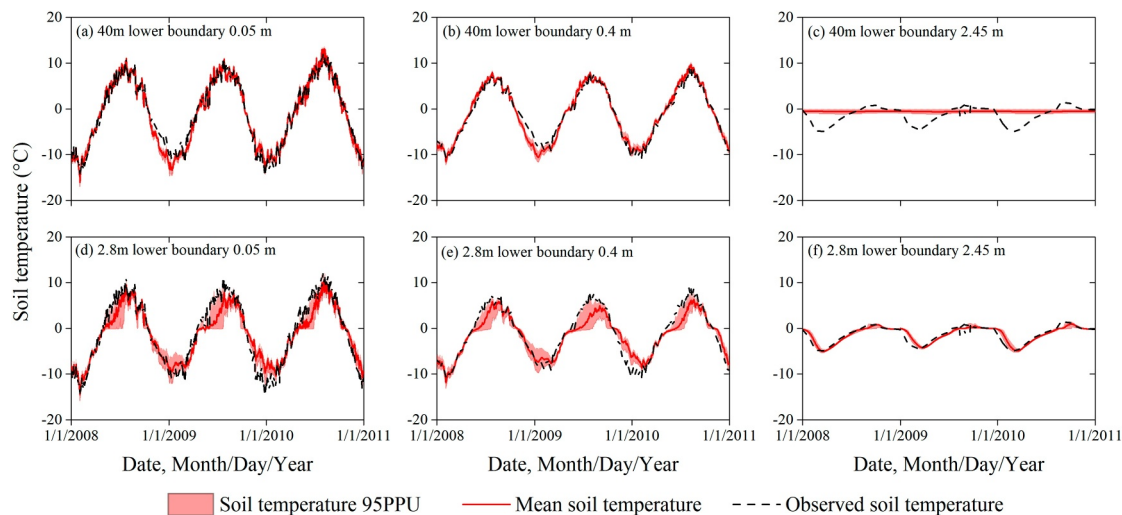


Figure 4. Simulated soil temperature (ST) at 0.05 m, 0.4 m, and 2.45 m depth at the Tanggula site (TGL), to demonstrate the Simultaneous Heat and Water model's sensitivity to two different lower boundary settings. The top row (a, b, and c) uses a stable temperature at 40 m depth estimated by an elevation-dependent regression as the thermal lower boundary, while the bottom row (d, e, f) uses the observed ST at 2.8 m as the thermal lower boundary. The shaded area represents the 95 percentile prediction uncertainty (95PPU) derived from 1,000 simulations with varied four key parameters. (c) Shows that a 40 m boundary causes substantial underestimation of seasonal temperature fluctuation at deep depth (2.45 m), a bias that persists across all parameter sets. The parameter disturbance test indicates that the bias arises from the boundary condition itself, not from parameter uncertainty. These simulations were based on the work of Zhao et al. (2023).

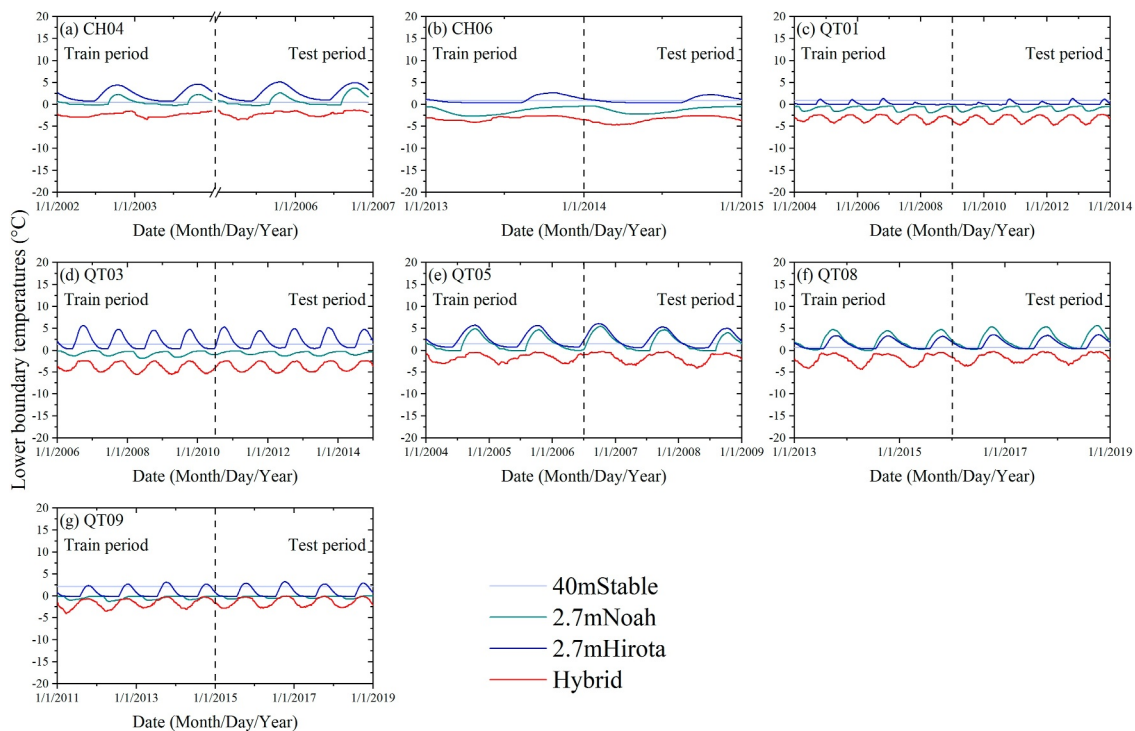


Figure 5. Comparison of the four lower boundary temperature scenarios estimated by different methods during the observation periods for each site. 40mStable: a fixed deep-ground temperature at a 40 m depth. 2.7mNoah: the uncorrected simulated soil temperature at 2.7 m depth from the Noah land surface model (LSM). 2.7mHirota: the temperature estimated by the Hirota method at 2.7 m. Hybrid: the Noah LSM simulation at 2.7 m depth after being corrected by the Random Forest model. For the Hybrid scenario, the vertical dashed line in each panel separates the training period (first 50% of data) from the testing period (second 50%).

2.2.3. Noah LSM Simulations and Correction for Lower Boundary Conditions

In this study, we applied simulations from the Noah LSM to generate high-quality lower boundary temperature and moisture content data for driving the SHAW model. The Noah LSM is a widely used one-dimensional LSM for simulating heat and water exchange and transfer processes across the atmosphere-canopy/snowpack-soil continuum. Despite its relatively simplified representation of soil heat and water dynamics, the Noah LSM can provide reasonable simulations of ST and water content in permafrost regions without precise lower boundary constraints (Wu et al., 2018), a capability attributed to its use of the Crank–Nicolson finite difference method. We applied a modified version of the Noah LSM developed by Ji et al. (2022) to simulate daily ST and total water content (liquid plus ice) at a depth of 2.7 m for each site from 1979 to 2018. These simulations served as the initial lower boundary conditions for the subsequent SHAW model simulations. The 2.7 m depth was selected because it is close to the average ALT of ~2.12 m on the QTP (Wang et al., 2022) and because the Noah LSM has been shown to provide relatively reliable simulations at this depth (Ji et al., 2022; Wu et al., 2018). Meteorological forcing data for the Noah LSM were obtained from the ITP-forcing data set. The Noah LSM soil column was discretized into 18 layers spanning 0–15.2 m, which includes the 2.7 m layer, following the configuration of Ji et al. (2022). The lower boundary of the Noah LSM was set at a depth of 40 m with a fixed temperature determined using an empirical relationship between ground temperature at 40 m and elevation (Chen et al., 2015). A free drainage boundary condition was applied for water migration. The Noah LSM parameters were adopted from Wu et al. (2018), which were previously calibrated using observations from permafrost sites on the QTP.

However, the initial Noah LSM simulations consistently overestimated soil temperatures at the 2.7 m depth. For instance, at CH04, QT05, and QT08, where the ALT is less than 2.7 m (Table 1), temperatures below this depth should theoretically remain below 0°C throughout the year. Yet, the Noah LSM consistently predicted temperatures above 0°C at these depths (see Section 3.1 and Figure 5). Despite these deviations, a consistent trend was observed between the Noah LSM estimates and the expected temperature behavior. This suggested that, after appropriate correction, the Noah LSM estimates of ST at 2.7 m could serve as suitable lower boundary conditions for the SHAW model.

Table 2
Vegetation Types and Key SHAW Parameters Used for the Seven Permafrost Monitoring Sites

Site	Vegetation type	Canopy height (m)	Leaf width (m)	Albedo (dimensionless)	Leaf area index (m ² /m ²)	Root depth (m)
CH04	Grassland	0.56	0.01	0.1	1.6	0.50
CH06	Barren	0.2	0.003	0.16	0.7	0.30
QT01	Grassland	0.56	0.01	0.1	1.6	0.50
QT03	Grassland	0.56	0.01	0.1	1.6	0.50
QT05	Grassland	0.56	0.01	0.1	1.6	0.50
QT08	Grassland	0.56	0.01	0.1	1.6	0.50
QT09	Barren	0.2	0.003	0.16	0.7	0.30

To address the Noah LSM's overestimation of soil temperatures at 2.7 m, an RF model was developed. The RF model utilized Noah LSM-simulated soil temperatures at depths of 0.02, 0.39, and 2.7 m as predictor variables. These depths were chosen because they correspond to the atmosphere-controlled near-surface layer, the intermediate layer influenced by both upper and lower boundary conditions, and the lower boundary-controlled layer, respectively, thereby effectively capturing the shifting thermal regime across the active layer. Recognizing that deep ST is influenced by cumulative downward heat fluxes from previous days, the input vectors for the RF model consisted of 30-day averages of these simulated temperatures. Due to the lack of observed soil temperatures at the lower boundary depth for model training, an optimization approach was employed to determine “true” lower boundary temperatures at each site. Specifically, 100 sets of potential lower boundary temperatures were generated for each site by decreasing the Noah LSM-simulated soil temperatures at 2.7 m in 0.1°C increments from 0.1°C to 10°C. These provisional boundary temperatures were evaluated by inputting them as lower boundary conditions for the SHAW model at each site and assessing the accuracy of the resulting SHAW simulations against observed values at various depths. The set of lower boundary temperatures that yielded the most satisfactory model performance was identified as the “true” lower boundary temperature series for each site. These series were subsequently used as the target variable for training the RF model. Only data within the observation periods of each site were used for model training, with the first 50% in time of the data used for training and the remaining 50% for testing. The trained RF model can then be used to correct lower boundary temperatures for any period and at any ungauged location within the QTP.

The hydraulic lower boundaries of the SHAW model at each site were set at the same 2.7 m depth using the total water content simulated by the Noah LSM. The SHAW model then automatically partitioned the ice and liquid water content based on the provided lower boundary temperature.

2.2.4. Model Parameters and Settings

The soil column of the hybrid modeling was stratified into 10 layers centered at 0.00, 0.02, 0.07, 0.13, 0.23, 0.39, 0.66, 1.1, 1.83, and 2.7 m that also serves as the lower boundary. The meteorological forcing data at each site were from the corresponding grid data provided by ITP-forcing. The initial conditions of ST and soil moisture profiles for SHAW were provided by the 1 January 1979 simulation of Noah LSM, which was initialized with a 500-year spin-up (Ji et al., 2022). The simulation spanned from 1979 to 2018 at a temporal resolution of 3h and a spatial resolution of 0.1°. Since all site observations began after 2000 (Table 1), the period from 1979 to 1999 was considered another spin-up period, which further eliminated any bias from the initial states.

The values of key vegetation parameters were identified by considering the parameter lookup tables from the Global Land Data Assimilation System (GLDAS), the NASA Land Data Assimilation System, and direct observations on the QTP (Li et al., 2011; Luo et al., 2004; Man et al., 2022). Soil parameter values were provided by the GLDAS lookup table, which is widely used to provide parameters for LSMs such as Noah-MP and CLM. The values of vegetation and soil parameters applied for each site are presented in Tables 2 and 3, respectively.

Table 3
Soil Texture Types and Key SHAW Parameters Used for the Seven Permafrost Monitoring Sites

Site	Soil type	Sand (%)	Silt (%)	Clay (%)	Bulk density (kg m ⁻³)	Saturated hydrological conductivity (cm h ⁻¹)	Saturated water content (m ³ m ⁻³)	Air entry potential (m)	Pore size distribution (dimensionless)	40 m depth soil temperature (°C)
CH04	Loam	43	39	18	1,390	1.22	0.439	−0.355	5.25	0.47
CH06	Sandy Loam	58	32	10	1,610	1.88	0.434	−0.141	4.74	0.84
QT01	Sandy Loam	58	32	10	1,610	1.88	0.434	−0.141	4.74	0.92
QT03	Sandy Loam	58	32	10	1,610	1.88	0.434	−0.141	4.74	1.41
QT05	Sandy Loam	58	32	10	1,610	1.88	0.434	−0.141	4.74	1.44
QT08	Loam	43	39	18	1,390	1.22	0.439	−0.355	5.25	0.61
QT09	Loam	43	39	18	1,390	1.22	0.439	−0.355	5.25	2.16

2.3. Performance Evaluations

To rigorously evaluate the performance of the proposed hybrid approach, we conducted an inter-comparison with four additional simulation scenarios. Three of the scenarios were conducted using different lower boundary settings to SHAW and the last one was direct simulation using Noah LSM. Together with the proposed hybrid approach, there are five scenarios for an inter-comparison of their ability to simulate active layer freeze-thaw processes:

1. Hybrid: This is the proposed hybrid approach, where the lower boundary was provided by Noah-LSM simulations at 2.7 m and corrected by a trained RF model.
2. 2.7mNoah: in this scenario, the uncorrected ST and moisture content series at a depth of 2.7 m from the Noah LSM were directly employed as the lower boundary conditions for the SHAW model simulations. As previously noted, the Noah LSM tends to overestimate ST at deep layers, which was expected to negatively affect the performance of the SHAW model.
3. 40mStable: this scenario represents a common approach for ungauged regions, setting the lower boundary at a 40 m depth with a stable temperature and zero water flux. This setting assumes that the ST and moisture are stable at great depths. We used an elevation-dependent empirical function from Chen et al. (2015) to estimate the ST at this depth:

$$T_{s,40m} = -0.00635x + 30.98 \quad (1)$$

where $T_{s,40m}$ is the ST at 40 m depth and x denotes the elevation (m a.s.l.). The estimated ST values are presented in Table 3.

4. 2.7mHirota: the SHAW model provides a simplified method for estimating the lower boundary temperature based on an extended force-restore method developed by Hirota et al. (2002). In this method, the lower boundary was set at 2.7 m, and the lower boundary ST at a given time step was determined by the temperatures of the lower boundary and the penultimate soil layer at the previous time step, as expressed by the equation:

$$T_{ns}^t = \begin{cases} T_{ns}^{t-1} + 1, & T_{ns}^{t-1} < T_{ns-1}^{t-1} \\ T_{ns}^{t-1} - 1, & T_{ns}^{t-1} \geq T_{ns-1}^{t-1} \end{cases} \quad (2)$$

where T_{ns}^t is the lower boundary temperature at a time step t , T_{ns}^{t-1} is the lower boundary temperature at the previous time step $t - 1$, and T_{ns-1}^{t-1} is the temperature of the penultimate soil layer ($ns - 1$), the closest soil layer to the lower boundary.

5. Noah: These are the direct simulations from the Noah LSM, using the model setup described in Section 2.2.3. Comparing the Hybrid scenario results to this one highlights the advantages of the hybrid framework over the standalone Noah LSM.

All five scenarios were evaluated for their performance at the seven sites. To ensure a fair inter-comparison, the four SHAW-based scenarios were run with the same input data and model parameters (Tables 2 and 3), isolating only the influence of the lower boundary condition.

We selected three representative depths at each site, that is, shallow (0–0.2 m), middle (approximately 0.5 m), and deep (below 1.0 m), for assessing simulation accuracy against observations. The Nash-Sutcliffe efficiency (NSE) coefficient was adopted as the primary evaluation criterion:

$$\text{NSE} = 1 - \frac{\sum_{i=1}^N (O^i - M^i)^2}{\sum_{i=1}^N (O^i - \bar{O})^2} \quad (3)$$

where O^i and M^i are the observed value and the simulated value at time i ; $\bar{O}$ is the mean of the observations over the entire period, and N is the total number of time steps. It is worth noting that because measurement depths varied slightly between sites, we compared observations with simulations at the closest available model depths.

3. Results

3.1. The Lower Boundary Conditions

We first compared the thermal lower boundaries generated by the four methods (Figure 5). In the 40mStable scenario, the thermal lower boundary temperatures, while different for each site, remain fixed throughout the simulation period. This is consistent with the real-world condition that soil temperatures at great depths (e.g., 40 m) are primarily influenced by geothermal heat and exhibit negligible interannual fluctuations. In contrast, the 2.7mNoah and 2.7mHirota scenarios both estimated temperatures consistently above 0°C at 2.7 m depth at most permafrost sites (Figure 5), a depth that is below the active layer at those sites. This contradicts the expected negative temperature in the permafrost body at these sites, indicating that the Noah LSM and Hirota methods overestimate soil temperatures at great depths.

The Hybrid scenario corrected the overestimation of the ST at the 2.7 m lower boundary depth observed in the 2.7mNoah scenario. According to our experiments, to achieve a good performance in the SHAW simulations of ST, the lower boundary temperatures at sites CH04, QT05, QT08, and QT09 needed to be reduced by 4–6°C from the Noah simulations (Figures A1a, A1e–A1g). Moreover, despite the Noah LSM simulations being consistently negative at CH06, QT01, and QT03 (Figures 5b–5d), a 2°C reduction was still required to achieve more satisfactory simulations (Figures A1b–A1d). This demonstrates that the overestimation of deep ST by the Noah LSM is a widespread issue across the QTP and that a correction is necessary. The “true” lower boundary soil temperatures identified at each site were then used as the target variables to train the RF correction model. The trained RF model showed high accuracy for both the training and testing data sets (Figure A2) and was then used to correct the lower boundary temperatures for each site over the full period. The corrected lower boundary temperatures, shown as the Hybrid scenario in Figure 5, present negative and fluctuating temperature series at all sites, which better represent the characteristics of permafrost than the lower boundaries from other scenarios.

3.2. Performance of the Noah-SHAW Hybrid Model

Figures 6 and 7 show the simulated soil temperatures and moisture contents, respectively, at the shallow, middle, and deep layers of the active layer across the seven sites under different lower boundary scenarios, with corresponding NSE values presented in Tables 4 and 5. Note that while observations (50%) from the training period were used to train the RF model, the testing period provides an independent validation of the hybrid model's performance. In simulating ST, the 2.7mHirota scenario yielded the worst performance with the lowest NSE value of 0.63 for all sites during the test periods. The average NSE values for the 2.7mNoah and 40mStable scenarios were 0.74 and 0.66, respectively, indicating a slight improvement compared to the 2.7mHirota scenario. The Hybrid scenario performs best, attaining the highest average NSE (0.81) in simulating active layer temperature.

The different scenarios produced similar simulated ST changes and close NSE values at shallow depths at each site (less than 0.2 m, first column of Figure 6). This demonstrates the excellent performance of the SHAW model in capturing near-surface thermal dynamics, regardless of the specific lower boundary thermal conditions. However, the performance of the 40mStable, 2.7mNoah, and 2.7mHirota scenarios worsened significantly in

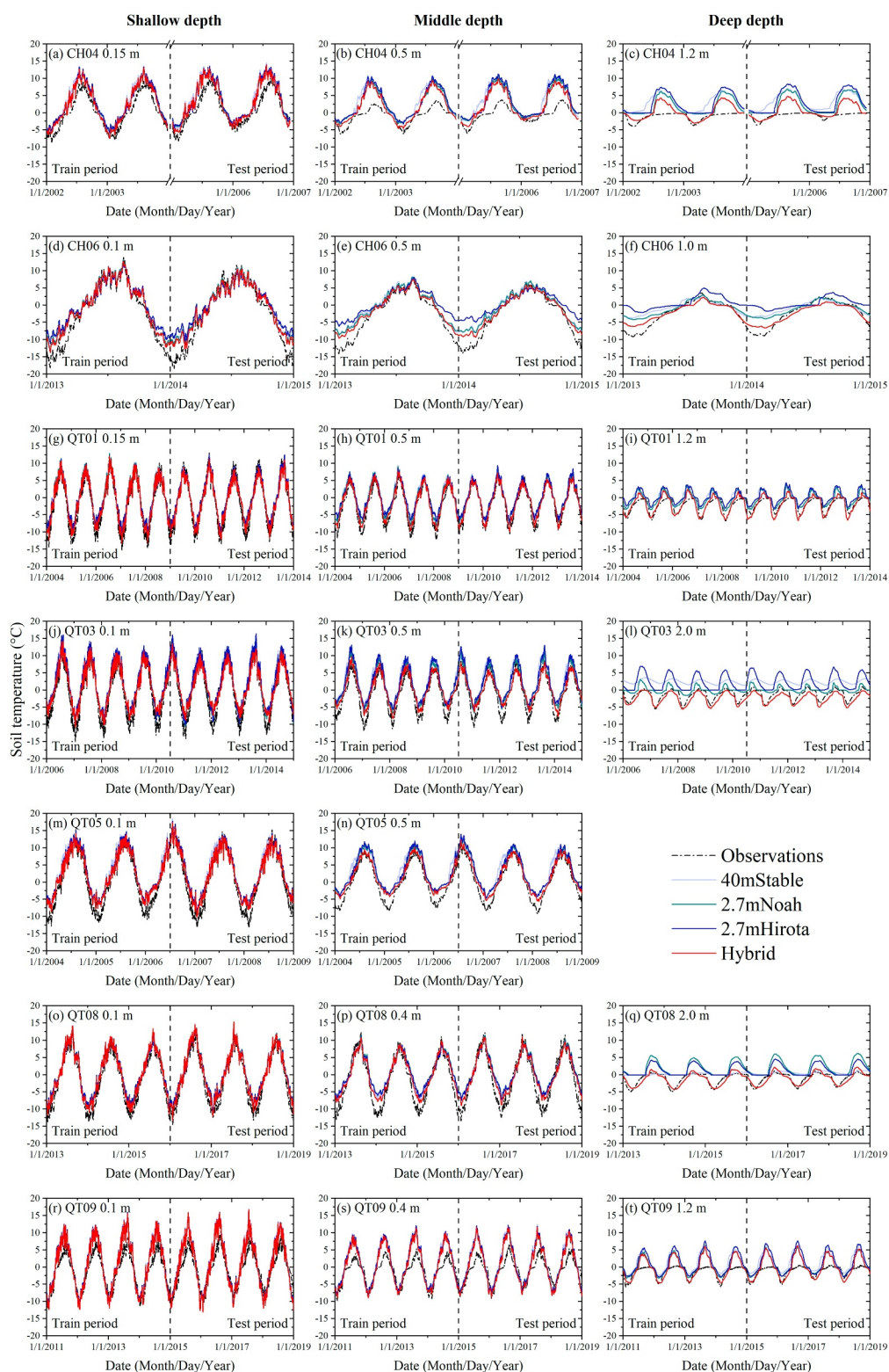


Figure 6. Simulated daily soil temperatures from the four simultaneous heat and water model scenarios at representative depths across the seven sites. Note that the deep active layer (0.9 and 2.1 m) comparison at QT05 was excluded due to a large mismatch between observation and simulation depths (over 20 cm far away from the closest simulated depths). The vertical dashed line separates the training period (first 50%) and testing period (second 50%) for the Hybrid scenario.

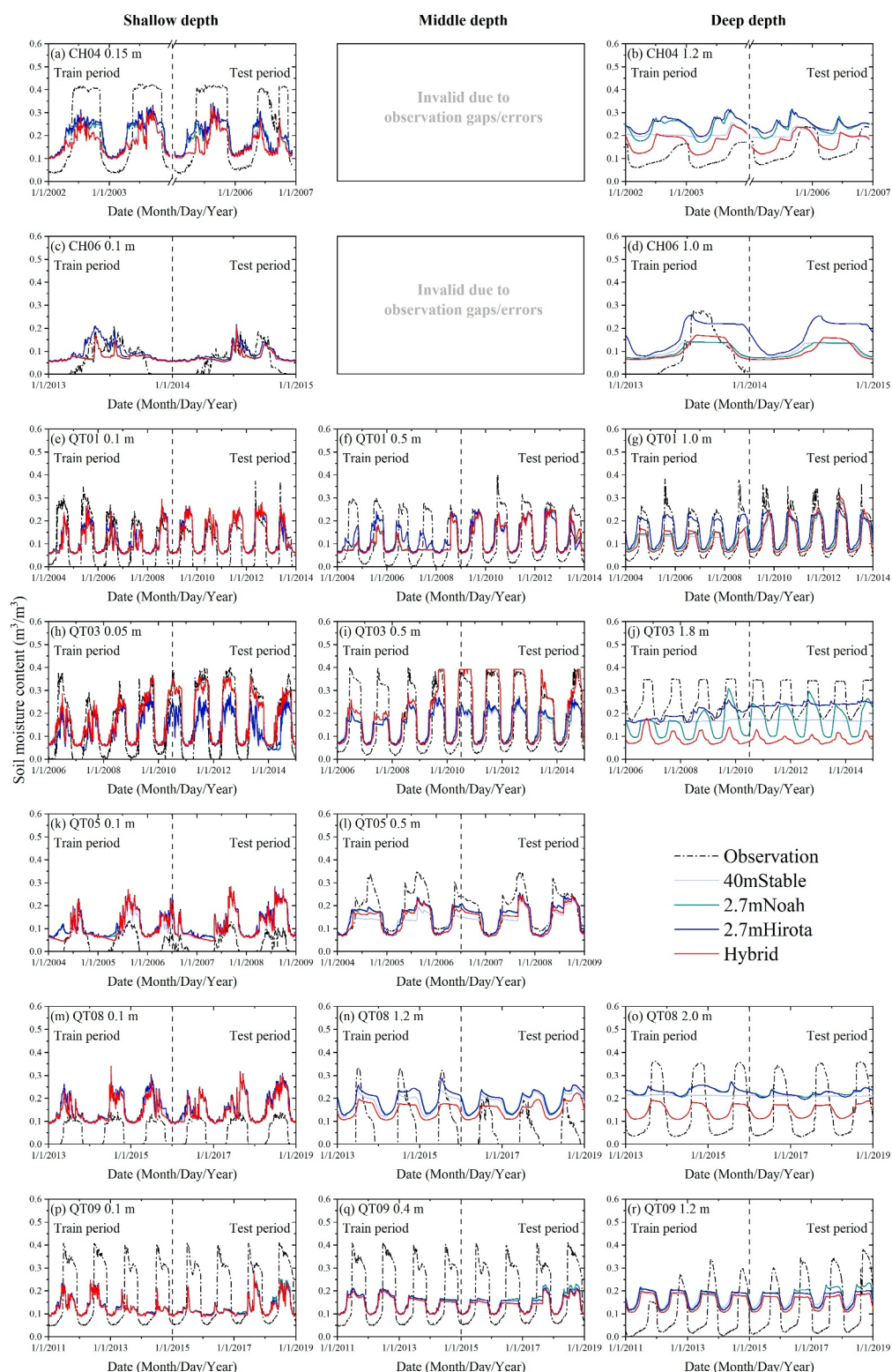


Figure 7. Simulated daily soil moisture contents from the four simultaneous heat and water model scenarios at representative depths across the seven sites. The vertical dashed line separates the training and testing periods for the Hybrid scenario. Note that some comparisons were excluded due to large mismatches between observation and simulation depths (deep layer at QT05) or significant errors and gaps in the observational data (middle layers at CH04, CH06, and QT08).

Table 4*Nash-Sutcliffe Efficiency (NSE) for Daily Soil Temperature Simulations From the Five Scenarios at Representative Depths*

Site	Depth	40mStable	2.7mNoah	2.7mHirota	Hybrid	Noah
CH04	0.15 m	0.48 (0.60)	0.64 (0.71)	0.58 (0.66)	0.65 (0.74)	0.40 (0.63)
	0.50 m	−4.08 (−2.48)	−2.79 (−1.44)	−3.78 (−2.09)	−1.86 (−0.95)	−4.08 (−2.02)
	1.20 m	−27.22 (−12.39)	−19.61 (−7.88)	−30.69 (−13.06)	−6.25 (−2.44)	−18.50 (−6.48)
CH06	0.10 m	0.88 (0.89)	0.89 (0.89)	0.79 (0.82)	0.90 (0.91)	0.95 (0.94)
	0.50 m	0.81 (0.80)	0.82 (0.82)	0.52 (0.58)	0.90 (0.89)	0.92 (0.89)
	1.00 m	0.52 (0.51)	0.61 (0.64)	−0.19 (−0.11)	0.88 (0.86)	0.86 (0.85)
QT01	0.15 m	0.93 (0.90)	0.92 (0.91)	0.92 (0.88)	0.94 (0.94)	0.91 (0.90)
	0.50 m	0.85 (0.76)	0.87 (0.80)	0.83 (0.71)	0.95 (0.92)	0.75 (0.76)
	1.20 m	−0.47 (−0.58)	0.06 (−0.07)	−0.81 (−0.81)	0.46 (0.54)	−1.45 (−0.59)
QT03	0.10 m	0.88 (0.82)	0.90 (0.86)	0.87 (0.83)	0.88 (0.88)	0.96 (0.98)
	0.50 m	0.55 (0.36)	0.75 (0.62)	0.57 (0.43)	0.83 (0.81)	0.95 (0.92)
	2.00 m	−5.18 (−5.89)	−0.25 (−0.61)	−5.79 (−5.00)	0.34 (0.59)	0.30 (0.17)
QT05	0.10 m	0.89 (0.85)	0.89 (0.86)	0.89 (0.85)	0.92 (0.89)	0.98 (0.96)
	0.50 m	0.72 (0.55)	0.78 (0.67)	0.75 (0.61)	0.91 (0.83)	0.90 (0.82)
QT08	0.10 m	0.91 (0.91)	0.89 (0.88)	0.90 (0.89)	0.93 (0.93)	0.72 (0.71)
	0.40 m	0.82 (0.83)	0.76 (0.75)	0.78 (0.76)	0.86 (0.88)	0.69 (0.70)
	2.00 m	−0.72 (−0.47)	−3.97 (−2.62)	−1.86 (−1.61)	0.69 (0.76)	−5.56 (−3.32)
QT09	0.10 m	0.51 (0.58)	0.60 (0.62)	0.56 (0.60)	0.62 (0.64)	0.35 (0.53)
	0.40 m	−0.21 (−0.05)	0.12 (0.15)	−0.06 (0.03)	0.15 (0.22)	−0.17 (0.04)
	1.20 m	−3.92 (−1.92)	−1.77 (−0.93)	−4.03 (−2.31)	−1.74 (−0.52)	−2.93 (−1.08)
All sites	All depth	0.66 (0.61)	0.74 (0.70)	0.63 (0.59)	0.81 (0.82)	0.69 (0.72)
	Shallow (<0.2 m)	0.85 (0.83)	0.86 (0.85)	0.84 (0.82)	0.87 (0.87)	0.83 (0.85)
	Middle (~0.5 m)	0.59 (0.53)	0.68 (0.63)	0.58 (0.52)	0.77 (0.76)	0.66 (0.67)
	Deep (>1.0 m)	−1.82 (−2.19)	−1.07 (−0.73)	−2.78 (−2.09)	0.10 (0.37)	−1.61 (−0.76)

Note. Values are for the test period, with NSE values for the training period shown in parentheses. Bold values indicate the highest NSE among all scenarios for a given site and depth.

simulating temperatures at the middle and deep active layers. Substantial overestimation of ST was observed at 0.5 m at CH04 and 0.4 m at QT09 during thaw periods for all scenarios (Figures 6b and 6s). Moreover, a slight overestimation of ST during the freeze period was evident at the middle depth CH06, QT03, QT05, and QT08 (Figures 6e–6k, 6n, and 6p). While these systematic overestimations may be partly attributed to the biases in the forcing data, the hybrid method (Hybrid) showed the least overestimation and the closest agreement with observations. Moreover, despite the overestimation of soil temperatures during thawing periods at CH04 and QT09, the hybrid method provided more accurate ST simulations of the freezing period at middle depth of CH04, QT01, and QT09 (Figures 6b–6h and 6s). Consequently, the mean NSE value of the hybrid method at middle depths (0.77) was significantly higher than those of other scenarios in the test periods (Table 4).

At deep soil depths (deeper than 1 m), the 40mStable, 2.7mNoah, and 2.7mHirota scenarios all substantially overestimated ST compared to the observations (third column in Figure 6) due to their overestimated lower boundary temperatures, leading to negative NSE values (the last row in Table 4). In contrast, the hybrid method, constrained by a corrected lower boundary, demonstrates a significant improvement in capturing temperature variations at deep soil depths across most sites (Figure 6) and obtains the highest NSE values (Table 4).

Assessing model performance for soil moisture is challenging due to significant uncertainties in the observational data. For instance, numerous unrealistic zero water content values were observed during freezing periods at CH06 (Figures 7c and 7d), QT05 (Figure 7k), and QT08 (Figures 7m and 7n). The measured moisture contents at shallow depths appear unrealistically lower than those at middle depths at QT05 and QT08 (Figures 7k–7n). Uncertainties from ITP-forcing precipitation data (Figure 2b) may also introduce biases in the simulation of soil

Table 5*Nash-Sutcliffe Efficiency (NSE) for Daily Soil Moisture Content Simulations From the Five Scenarios at Representative Depths*

Site	Depth	40mStable	2.7mNoah	2.7mHirota	Hybrid	Noah
CH04	0.15 m	0.10 (0.33)	0.32 (0.50)	0.32 (0.52)	0.09 (0.32)	0.33 (0.32)
	1.20 m	−1.04 (−6.77)	−2.59 (−11.62)	−3.17 (−14.40)	0.11 (−2.49)	−3.32 (−13.12)
CH06	0.10 m	0.23 (0.24)	0.23 (0.24)	0.25 (0.17)	0.25 (0.25)	−0.06 (−0.51)
	1.00 m	NA (0.39)	NA (0.44)	NA (−0.18)	NA (0.61)	NA (−0.29)
QT01	0.10 m	0.39 (0.40)	0.39 (0.41)	0.36 (0.48)	0.41 (0.43)	−0.67 (−0.05)
	0.50 m	0.58 (0.11)	0.63 (0.12)	0.56 (0.37)	0.69 (0.16)	0.16 (0.07)
	1.00 m	0.67 (0.29)	0.75 (0.44)	0.68 (0.53)	0.81 (0.51)	0.26 (−0.07)
QT03	0.05 m	0.43 (0.54)	0.48 (0.55)	0.48 (0.54)	0.82 (0.66)	0.45 (−0.08)
	0.50 m	0.47 (0.45)	0.54 (0.48)	0.52 (0.46)	0.77 (0.57)	0.43 (0.16)
	1.80 m	−1.87 (−1.34)	−0.81 (−1.41)	−0.11 (−0.43)	−6.44 (−4.70)	0.68 (0.67)
QT05	0.10 m	−4.41 (−0.69)	−5.76 (−1.55)	−5.79 (−1.66)	−4.73 (−1.06)	−17.77 (−8.06)
	0.50 m	0.28 (0.05)	0.62 (0.50)	0.62 (0.52)	0.48 (0.41)	0.83 (0.76)
QT08	0.10 m	−3.27 (−2.98)	−3.66 (−3.68)	−3.62 (−3.67)	−2.91 (−2.61)	−5.18 (−6.94)
	1.20 m	−3.52 (−0.16)	−4.05 (−0.47)	−3.93 (−0.44)	−1.90 (0.11)	−5.39 (−0.92)
	2.00 m	−0.21 (−0.21)	−0.21 (−0.31)	−0.17 (−0.31)	0.32 (0.35)	−0.12 (−0.11)
QT09	0.10 m	−0.01 (−0.01)	0.04 (0.01)	0.01 (0.01)	−0.01 (0.00)	−0.25 (0.08)
	0.40 m	0.13 (0.18)	0.25 (0.23)	0.20 (0.24)	0.12 (0.20)	0.37 (0.38)
	1.20 m	0.07 (−0.40)	0.22 (−0.32)	0.18 (−0.39)	0.24 (−0.12)	0.20 (−0.34)
All sites	All depths	0.29 (0.26)	0.35 (0.27)	0.33 (0.30)	0.35 (0.29)	0.17 (0.10)
	Shallow (<0.2 m)	0.33 (0.36)	0.36 (0.37)	0.35 (0.38)	0.44 (0.39)	0.04 (−0.04)
	Middle (~0.5 m)	0.38 (0.31)	0.48 (0.38)	0.45 (0.43)	0.51 (0.37)	0.36 (0.30)
	Deep (>1.0 m)	0.11 (0.04)	0.17 (−0.04)	0.13 (−0.00)	0.02 (0.04)	0.08 (−0.02)

Note. Values are for the testing period, with NSE values for the training period shown in parentheses. For site CH06, testing period results at the deep layer are marked as “NA” (Not Available) due to significant errors in the observational data. Bold values indicate the highest NSE among all scenarios for a given site and depth.

water content (SWC). Although these uncertainties could negatively affect the evaluation of model performance, the inter-comparison across scenarios remains reliable because each scenario faced identical conditions. However, such treatment lowered the calculated performance metrics, limiting their direct comparability with values reported in other studies. In addition, we have also excluded the observations with serious missing values or errors from the validation, such as those at the middle depths of CH04, CH06, QT08 (Figure 7).

Many scenarios had negative NSE values for soil moisture simulations at specific depths, reflecting the observational uncertainties. The Hybrid scenario has the highest mean NSE value of 0.35 across all sites in the test periods, followed by the 2.7mNoah scenario (also 0.35, but with a lower NSE in the train period), 2.7mHirota (0.33), and 40mStable (0.29). For the shallow and middle soil layers, the SHAW-based scenarios (2.7mNoah, 40mStable, and Hybrid) displayed similar patterns, underestimating liquid moisture content during summer at CH04 and QT09 (Figures 7a–7p and 7q), whereas overestimating it at QT05 and QT08 (Figures 7k and 7m). The Hybrid scenario showed advantages over other scenarios at the deep active layer of most sites, with the notable exception for QT03. At a depth of 1.8 m in QT03, the simulated soil temperatures of the Hybrid scenario were consistently below 0°C (Figure 6l), resulting in incomplete melting of simulated soil ice content and significantly lower liquid content compared to the other scenarios and to the observations (Figure 7j). Overall, despite a few such errors at individual site depths like 0.5 m at QT03, the hybrid method (Hybrid) demonstrates the highest NSE values in both training and test periods over other scenarios at most depths. The more reasonable lower boundary thermal conditions applied in the Hybrid scenario significantly improve the SHAW model's simulation of phase change and the distribution of liquid and ice content within the soil.

3.3. Comparison With the Noah LSM Simulations

The hybrid approach we proposed also has advantages in simulating active layer hydrological and thermal processes when compared against the standalone Noah LSM. The soil temperatures simulated by our hybrid approach (Hybrid) generally show better agreement with observations than those from the Noah LSM (Noah) at most depths of the representative sites, achieving a higher overall NSE value (0.81 for the hybrid approach vs. 0.69 for Noah at all sites during test periods, as presented in Table 4). Benefiting from the use of more accurate lower boundary temperature, the hybrid approach is particularly advantageous in simulating the ST at the deep active layer, in comparison with the systematic overestimations from the Noah LSM at these depths (the third column in Figure 8). Consequently, the average NSE value of simulated ST at the deep active layer for all sites from the hybrid approach (0.10) is much higher than those from Noah (−1.61). At some specific sites such as QT01, QT03, and QT08 (Figures 8i–8l and 8q), the advantages of the hybrid approach are particularly pronounced. The simulated temperatures from the hybrid approach also align more closely with the observations than those from Noah at shallow and middle depths in most sites (Figures 8a–8d, 8e, 8g, 8h, 8o and 8p). The Noah LSM tends to overestimate the peak temperature during summer and underestimate it during winter. However, the hybrid approach also exhibits some biases, such as overestimating the winter ST, and presents lower NSE values than Noah at the shallow depths of QT03 and QT05 (Figures 8j, 8k, and 8n).

The overall NSE value for unfrozen water content simulations from our hybrid model (0.35) is substantially higher than that from the Noah model (0.17). The hybrid approach provides a closer approximation to observed unfrozen water content during the freezing periods (especially at QT01 and QT03 as presented in Figure 9), suggesting that the robust representation of active layer hydrology in the SHAW model is better suited for simulating freeze-thaw processes within permafrost regions. Conversely, Noah exhibits a notable overestimation of residual unfrozen water content during the freezing periods across most sites (Figure 9), which can be attributed to the inherent limitations of its parameterizations in representing the distribution of ice and liquid water, as well as the phase change process. Therefore, although the hybrid approach performs more poorly than the Noah LSM at specific sites (Figures 9j and 9l), it exhibits a very significant improvement over the Noah LSM both in simulating unfrozen water content curves during frozen seasons.

Overall, by combining the advanced soil freeze-thaw parameterizations of the SHAW model with high-quality lower boundary constraints, the hybrid modeling approach developed in this study demonstrates substantially improved accuracy compared to the Noah LSM in simulating both the ST and unfrozen water content in the permafrost active layer.

4. Discussion

4.1. Strengths of Heat-Water Coupled Modeling for Permafrost Simulation

While LSMs are valuable tools for simulating permafrost dynamics, significant disparities exist in their capabilities to accurately represent heat and water transport within the active layer. These discrepancies arise primarily from differences in their process parameterization schemes and input data (Andresen et al., 2020; Burke et al., 2020; Slater & Lawrence, 2013). Many widely used models lack comprehensive consideration of complex heat and water transfer processes within frozen ground, such as non-conductive heat transfer and frozen ground hydrology. Consequently, their ability to accurately simulate temperature, freeze-thaw cycles, and ice content within permafrost is limited (Schädel et al., 2024). For instance, our results (Figures 8 and 9) and previous studies have demonstrated significant biases in Noah LSM simulations of permafrost temperature and unfrozen water content (Letcher et al., 2024; Zheng et al., 2017). Similarly, the CLM has been shown to exhibit substantial biases in unfrozen water content simulations, partly attributed to its inadequate soil matric potential scheme (Xiao et al., 2013). Moreover, most widely used LSMs are considered heat and water “uncoupled” models due to their incomplete consideration of the intricate interactions between heat and water during the freeze-thaw cycle (Hu, Zhao, et al., 2023; Yu et al., 2018).

The SHAW model stands out due to its comprehensive consideration of coupled heat and water processes. It incorporates detailed heat transfer mechanisms within the active layer, including conductive heat transfer, latent heat, and convective heat transfer resulting from liquid/vapor flow (Flerchinger, 2000; Hu, Zhao, et al., 2023). These thorough representations of heat transport enable the SHAW model to accurately capture the thermal

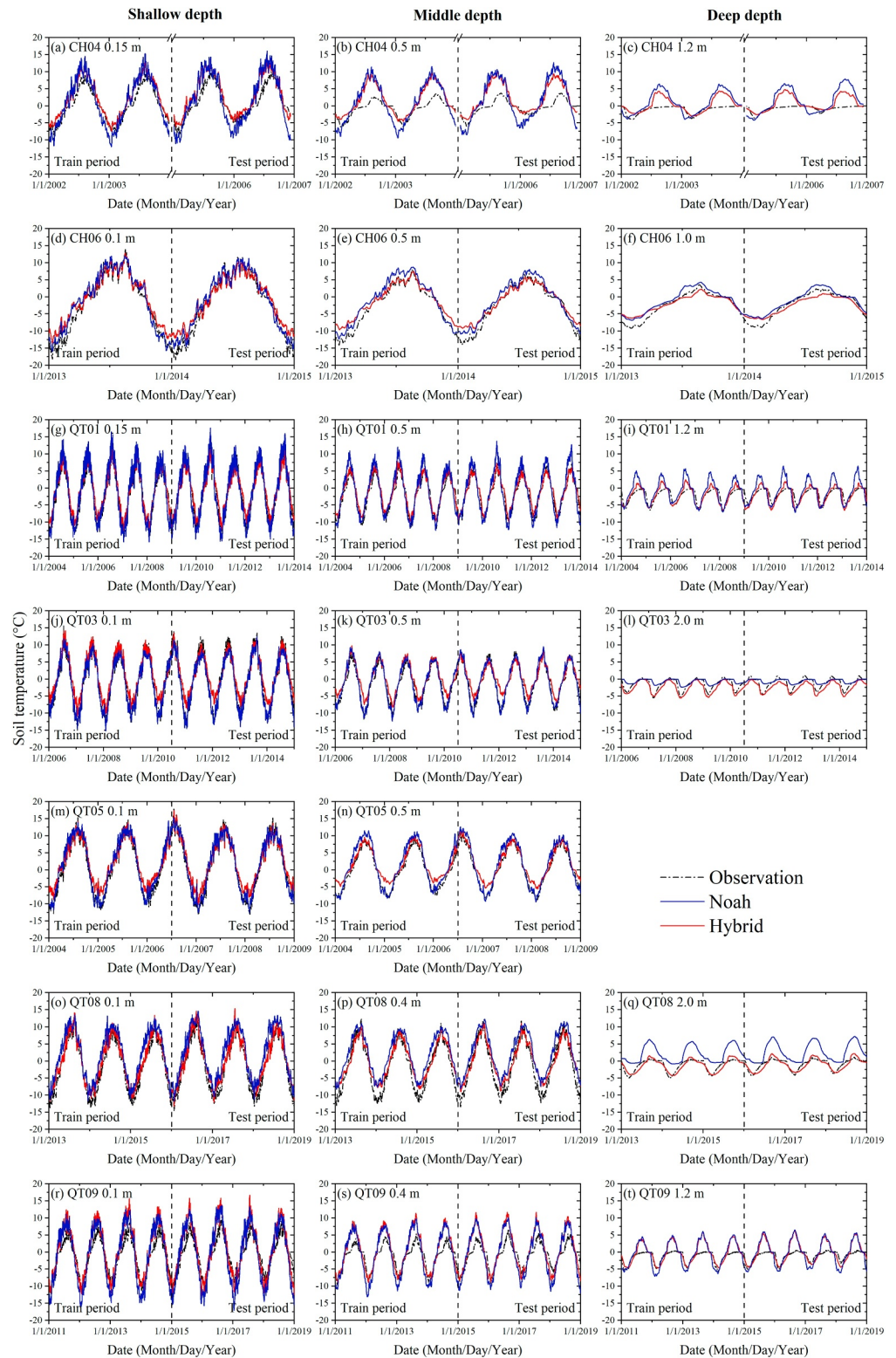


Figure 8. Comparison of daily soil temperatures simulated by the hybrid approach and the standalone Noah land surface model against observations at representative depths of the seven sites. The vertical dashed line separates the training and testing periods for the Hybrid approach's Random Forest model.

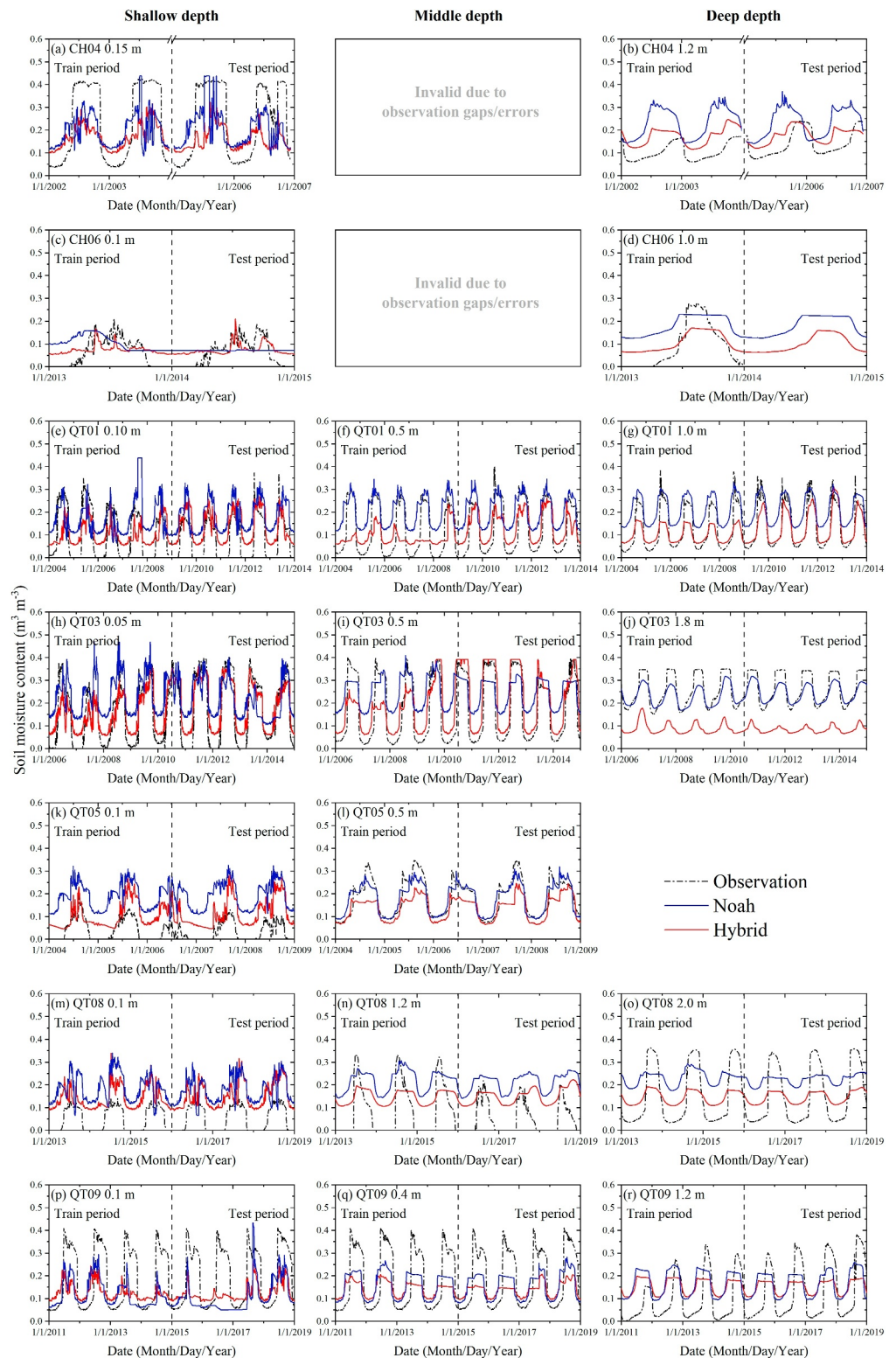


Figure 9. Comparison of daily soil moisture contents simulated by the hybrid approach and the standalone Noah land surface model against observations at representative depths of the seven sites. The vertical dashed line separates the training and testing periods. Note that comparisons at the middle depths of CH04, CH06, and QT08 were excluded due to significant errors and gaps in the observational data.

regimes of frozen soil, particularly the influence of latent heat and convective heat (Zhao et al., 2022). Furthermore, the SHAW model incorporates a hydraulic conductivity scheme that reduces soil permeability as pore ice content increases, allowing for the slow movement of unfrozen water between unfrozen soil layers and partially frozen layers. This study utilized an improved version of the SHAW model incorporating an enhanced soil matric potential scheme that more accurately simulates unfrozen water content by considering the effects of pore ice (Zhao et al., 2023). This results in significantly higher accuracy in predicting changes in soil moisture content during the freezing periods.

However, the SHAW model still exhibited limitations in simulating the peak values of SWC during thaw periods. Interestingly, similar limitations were observed in the Noah LSM simulations, suggesting that current numerical simulation methods still face challenges in accurately representing hydrological processes. These challenges may be attributed to both the inadequate representation of hydrological processes within the models and significant uncertainties arising from input data and observations (Figure 2b). Future research should focus on: (a) developing more advanced soil hydrology parameterizations and (b) obtaining higher-precision forcing data.

4.2. Role of Lower Boundary Condition in Permafrost Modeling

Lower boundary conditions play a crucial role in all LSMs by constraining the exchange of heat and water at the bottom of the simulated soil profile (Decharme et al., 2013; Hermoso De Mendoza et al., 2020). However, the degree to which the lower boundary condition influences the simulation of near-surface active layer dynamics varies significantly between models (Alkhaier et al., 2012; Chen et al., 2019; Cui et al., 2020; Hermoso De Mendoza et al., 2020), a distinction rooted in their fundamental design philosophies and numerical methods. This difference is particularly consequential for simulating the high-temperature, thermally sensitive permafrost of the QTP (Hu, Ma, et al., 2023).

Many widely-used LSMs, including Noah, are designed for operational atmospheric coupling and employ a sequentially coupled framework. In this approach, the governing equations for heat and water are solved in sequence (Yu et al., 2018). For instance, the soil heat equation is typically solved first for the entire soil column, often using a numerically stable implicit scheme like the Crank-Nicolson finite difference method, and the resulting new temperature profile is then used to solve for soil hydrology. This design, combined with a typically coarse vertical resolution, has a practical consequence: it numerically damps or buffers the influence of the deep lower boundary condition on the surface layers. As a result, these models are relatively insensitive to the precise state of a deep boundary, which is why the common practice of setting a fixed temperature at a great depth (e.g., 40 m) is often considered an acceptable solution for them.

In contrast, the SHAW model is designed to be a tool for capturing detailed soil physics. It employs a fully implicit, fully coupled framework, which is necessary to resolve the sharp non-linearities of soil freeze-thaw processes. SHAW uses a Newton-Raphson method that adaptively subdivides the time step, iteratively solving for the entire soil profile as a single, interconnected system until the changes in ST or moisture content between steps converge to a predefined threshold (typically 0.001) (Flerchinger, 2000). The consequence of this tightly integrated design, however, is that the lower boundary is not a distant, numerically damped input, but an immediate and integral constraint on the iterative solution for the entire system. Because the Newton-Raphson method constructs and solves a system based on a Jacobian matrix that links all nodes, the lower boundary's state directly informs the solution across the entire profile at every iteration. This tight coupling amplifies the influence of the boundary on the overlying layers. Its numerical precision and detailed vertical resolution mean that it will accurately resolve and propagate any signal, including the error from a lower-quality lower boundary condition, upward through the soil column with high fidelity. A simpler model like Noah, with its inherent buffering properties, is less ambitious in its physical representation and therefore less penalized by a crude boundary input.

The results of this study provide clear evidence of this theoretical sensitivity. When driven by a simplified, deep lower boundary (Figure 4c and the 40mStable scenario in Figure 6), the SHAW model failed to reproduce seasonal temperature fluctuations at depth, a failure not observed to the same degree in the less sensitive Noah LSM. Moreover, when constrained by shallow but low-quality lower boundary conditions (the 2.7mNoah and 2.7mHirota scenarios in Figure 6), the model also propagated these errors, leading to significant biases in the

active layer simulations. This demonstrates that the model is sensitive not only to the depth of the boundary but also its physical accuracy. We therefore recommend that a shallow and precise lower boundary condition be used to obtain accurate permafrost simulations from such using coupled heat-water models like SHAW.

However, determining reasonable lower boundary conditions for permafrost simulations presents a significant challenge, particularly for large-scale applications. The scarcity of observation sites across the QTP hinders the acquisition of sufficient deep-ground data. In current applications of LSMs, the lower boundary is often set at a considerable depth, assuming a fixed temperature or energy flux inferred from air temperature or the geothermal gradient (Hermoso De Mendoza et al., 2020; Wu et al., 2018). This simplified treatment often introduces systematic biases and fails to capture the temporal variations of energy and water exchange at the lower boundary. Even for LSMs that are relatively insensitive to their lower boundary conditions, such as the Noah LSM and CLM, these inadequate conditions can still degrade the performance of deep soil simulations (Decharme et al., 2013; Hermoso De Mendoza et al., 2020; Wu et al., 2018). The effect is evident in this study, where the Noah LSM systematically overestimates temperature in the deep active layer (>1 m; Figure 8). This bias is likely a combined result of the simplified lower boundary condition imposing a long-term systematic error and the model's uncoupled heat-water parameterizations, which cannot accurately represent coupled freeze-thaw physics.

This study proposes a novel approach that leverages outputs from existing LSMs and machine learning methods to generate reasonable lower boundary conditions for driving the SHAW model in ungauged areas of the QTP. This method effectively integrates the complementary strengths of SHAW and Noah LSM, achieving more accurate and physically consistent simulations of ST, moisture, and pore ice variations within the active layer. We suggest that this hybrid technique could similarly enhance other advanced models, offering a path toward more accurate permafrost simulations in data-sparse regions like the Arctic.

4.3. Limitations

This study has certain limitations. The hybrid model did not always accurately capture the peak moisture content during thaw periods. As a one-dimensional model, the SHAW model does not adequately account for the impacts of lateral flow within the active layer on soil hydrology. Lateral flow can significantly influence soil hydrology and the thermal regime of permafrost, and its omission in the model may contribute to inaccurate simulations (Kurylyk & Hayashi, 2016; Walvoord & Kurylyk, 2016). Moreover, the use of 10 km gridded forcing data may introduce discrepancies between the modeled results and in situ observations. The ITP-forcing data set is generated based on observations from national meteorological stations. While this ensures a close resemblance between the forcing data and observations for grid cells containing meteorological stations, discrepancies can arise in other grid cells, especially for precipitation, which is highly spatially heterogeneous in high-altitude regions. These discrepancies can ultimately hinder the ability of the proposed modeling approach to perfectly reproduce observed values at these specific sites. However, as demonstrated in this study, by effectively constraining the lower boundary, the hybrid modeling approach still achieved good accuracy in simulating soil temperatures and moisture dynamics, and it outperformed traditional modeling approaches in this regard. We expect that future improvements in the quality and resolution of climate forcing data will further enhance the performance of our hybrid approach.

This study utilized data from seven sites along the Qinghai-Tibet Highway, representative of alpine permafrost conditions in the central QTP. However, the QTP exhibits significant spatial variability, and the selected sites may not fully capture the diverse conditions present in areas with distinct hydrological and thermal characteristics. Further evaluation of the performance of our hybrid modeling approach in other data-limited areas is necessary. In addition, the widespread occurrence of abrupt permafrost thaw on the QTP (i.e., thaw slumps, thermokarst lakes, and active layer detachment) can substantially alter the thermo-hydrological conditions of frozen ground, thereby disturbing the landscape and threatening ecosystem and infrastructure stability (Luo et al., 2022; Xia et al., 2024). Current LSMs lack explicit process representations for abrupt thaw features, making the accurate simulation of permafrost dynamics under such conditions a persistent challenge (Schädel et al., 2024; Yi et al., 2024). Future development of permafrost models should therefore prioritize improving their applicability and robustness in complex, data-sparse regions affected by abrupt thaw.

5. Conclusions

This study presents a novel hybrid modeling approach for accurately simulating the thermo-hydrological dynamics of the active layer in permafrost regions on the QTP. The approach integrates the advanced physics of the SHAW model, high-quality meteorological forcing data, and a robust lower boundary condition determined through a combination of Noah LSM simulations and an RF correction. The performance of this hybrid approach was evaluated at seven representative permafrost sites on the QTP by comparing its simulations against those from the SHAW model with alternative lower boundary settings and from the standalone Noah LSM.

Results demonstrate that the hybrid approach significantly outperforms these alternative scenarios, particularly in simulating both ST and moisture content within the permafrost active layer. This enhanced performance can be attributed to two key factors: the advanced representation of freeze-thaw processes in the SHAW model and the well-constrained lower boundary conditions. The SHAW model incorporates comprehensive representations of coupled heat and water transfer processes within frozen ground, leading to more accurate simulations of the thermal regime compared to simpler models. The innovative approach of combining Noah LSM simulations with an RF correction effectively constrains heat transfer at the bottom of the active layer. This leads to more accurate simulations of temperature and unfrozen water content, particularly at deeper layers, where traditional lower boundary settings often introduce significant biases.

The hybrid approach also outperforms the Noah LSM, a widely used model for permafrost studies on the QTP, across various metrics. The mean NSE values for simulated ST and moisture content across the seven sites were 0.81 and 0.35 in the test periods, respectively, for the hybrid approach, significantly higher than those obtained by the Noah LSM (0.69 and 0.17). Notably, the hybrid approach provides more realistic simulations throughout the freeze-thaw cycle, correcting the Noah LSM's tendency to overestimate temperature at deeper layers and unfrozen water content during freezing seasons.

This proposed hybrid modeling approach enables the generation of long-term, large-scale, and accurate simulations of permafrost active layer dynamics across ungauged regions of the QTP. These high-quality simulations of permafrost temperature and moisture content are crucial for studying environmental and hydrological changes, assessing infrastructure risks, and quantifying soil carbon emissions associated with permafrost degradation.

Appendix A

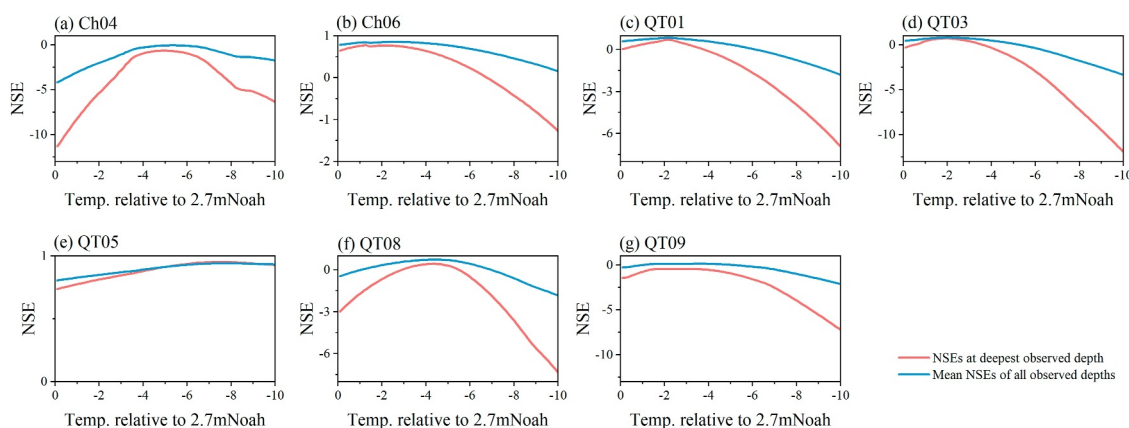


Figure A1. Results of the optimization procedure used to identify the “true” lower boundary temperature for Random Forest (RF) model training. The x-axis represents the temperature adjustment applied to the original 2.7 m Noah land surface model simulation, and the y-axis is the resulting nash-sutcliffe efficiency of the simultaneous heat and water model’s soil temperature simulation. The peak of each curve identifies the optimal temperature adjustment for that site. The temperature series corresponding to this peak was selected as the “true” lower boundary temperature and used as the target variable for training the RF model.

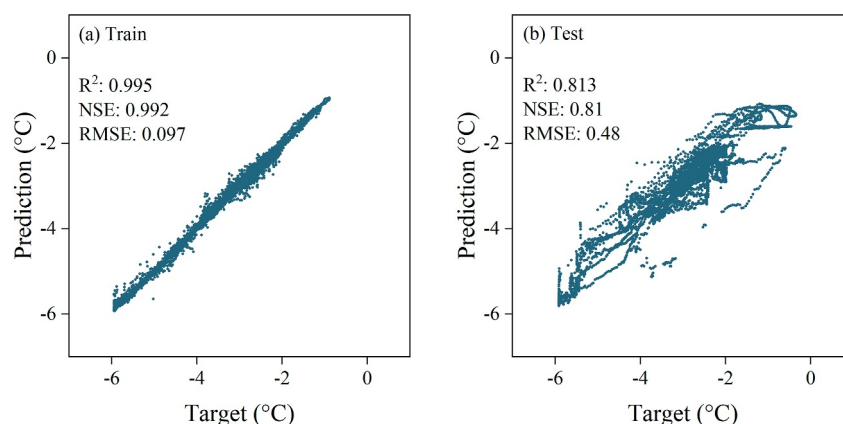


Figure A2. Performance of the Random Forest (RF) correction model. y-axis shows RF-predicted soil temperature at the lower boundary and x-axis shows the “true” lower boundary temperature for (a) the training data set and (b) the testing data set. The training set consists of the first 50% of the time-series data from all sites, and the testing set consists of the remaining 50%.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The source code for the original Simultaneous Heat and Water (SHAW) model version 3.1.2 is available on the USDA Agricultural Research Service website (USDA, 2019). The code for the improved SHAW model used in this study is publicly available at figshare (Zhao, 2022). The measured ST and moisture content data from the seven representative permafrost monitoring sites can be downloaded from the National Tibetan Plateau Data Center (Zhao et al., 2021). The codes and simulation results specific to this study are publicly available in a figshare repository (Zhao, 2025).

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